**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 012 · Computational Thinking — ADA

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does a higher clock speed mean?

Q2. Where is CPU cache?
A. On a remote server
B. Inside an optical disc
C. In a backup folder
D. In or near the CPU

Q3. Why is secondary storage needed?

Q4. A school uses DVDs for read-only videos and an HDD for a large archive. Correct both claims. Explain why. (a) DVDs are magnetic. (b) HDDs use only flash chips and have no moving parts.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

abstraction — Keeping the important details of a problem while ignoring irrelevant information.

decomposition — Breaking a complex problem into smaller, manageable parts.

algorithmic thinking — Planning a logical, ordered sequence of steps to solve a problem.

algorithm — A precise sequence of steps or instructions to solve a problem.

IPO — Input–Process–Output: a planning tool for what data goes in, what is done with it and what result comes out.

## New learning

Abstraction: Focus on the information needed to solve the problem and ignore irrelevant detail. For a journey-time estimate, distance and speed matter; car colour does not.

Decomposition: Break a complex problem into smaller, manageable parts that can be designed and checked separately, such as collecting inputs, calculating a result and displaying it.

Algorithmic thinking: Plan precise, logical steps in the correct order so that the problem can be solved. Decide what happens first, what must be checked and how the answer is produced.

IPO helps organise the steps: Input–Process–Output (IPO) is a useful way to plan the parts of a solution: gather the needed data, process it and show the result. IPO supports decomposition; it is not a replacement for algorithmic thinking.

Put ADA together: Journey-time calculator: abstract to distance and speed; decompose into input, checking, calculation and output; then order the steps to calculate and display hours.

## Worked example

Plan a journey-time calculator using abstraction, decomposition, algorithmic thinking and IPO.

1. ABSTRACTION: retain distance and speed; ignore details such as car colour that do not affect this estimate.

2. DECOMPOSITION: break the task into getting inputs, checking the speed, calculating time and displaying the answer.





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Q3. Input overdue days [1] and daily fine [1]; then multiply days by the daily fine [1]; finally display the amount due [1]; decompose the program into separate input, calculation and output tasks [1].

## Exit Ticket · Model answer

True

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. More clock cycles per second. This may let the CPU process more instructions if other factors are similar.

Q2. D — In or near the CPU

Q3. It keeps programs and files when power is off and usually provides lots of storage space.

Q4. (a) A DVD uses optical storage, read by a laser. (b) An HDD uses magnetic storage on spinning platters. Moving parts and portability affect which is suitable.

### I Do / We Do pages · Guided practice

Q1. Input: collect quantity and meal price; process: multiply quantity by price; output: display the total cost. Each stage is a smaller part of the problem.

Q2. Keep the number of meals and price per meal. Ignore plate colour because it does not affect the calculation.

Q3. 1. Input quantity = 3 and price = £4. 2. Process: multiply 3 × 4. 3. Output: display £12.

### You Do pages · Code First

```
files = 3
sizeMB = 4
totalMB = files * sizeMB
print(totalMB)
```

Expected result: 12

### You Do pages · Independent questions

Q1. Breaking a complex problem [1] into smaller, manageable problems [1].

Q2. Retain lesson times and rooms [1]; ignore irrelevant details such as wall colours [1]; this simplifies the problem without removing information needed to produce the timetable [1].





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3. IPO: input distance = 60 km and speed = 30 km/h; process distance ÷ speed; output journey time in hours.

4. ALGORITHMIC THINKING: in order, collect the values, check speed is greater than zero, divide 60 by 30, then display the result.

5. OUTPUT: 2 hours, assuming constant speed. Each ordered step has a clear purpose and can be checked.

## We Do · Practise with guidance

### Guided 1

Use IPO to decompose a lunch-cost calculator into three manageable stages.

Worked method: Input: collect quantity and meal price; process: multiply quantity by price; output: display the total cost. Each stage is a smaller part of the problem.

### Guided 2

For a lunch-cost calculator, which details must you keep, and which can you ignore through abstraction?

Worked method: Keep the number of meals and price per meal. Ignore plate colour because it does not affect the calculation.

### Guided 3

Three meals cost £4 each. Give the algorithm in the correct order, identifying input, process and output.

Worked method: 1. Input quantity = 3 and price = £4. 2. Process: multiply 3 × 4. 3. Output: display £12.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Use the IPO plan to finish this storage calculator: print the total storage needed in MB.

```
files = 3
sizeMB = 4
totalMB = files * sizeMB
```

write

### Question 1

Define decomposition. [2 marks]

### Question 2

Explain abstraction in a school timetable application. [3 marks]

### Question 3

Describe the steps of an algorithm for an overdue-fine calculator using IPO. State two inputs, the process, the output and one way to decompose the task. [5 marks]

## Exit Ticket · Answer before checking

Algorithmic thinking means arranging steps in a logical order to solve a problem.



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